

STANLEY CHUKWUEBUKA OKONKWO

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EDUCATION

MSc, Ecology and Nature Management | Peoples' Friendship University of Russia (RUDN University) 2024 — 2026

Thesis: Ecological Impacts of Urbanisation on Green Spaces and Drainage Patterns in Lagos State, Nigeria

BSc, Geological Sciences | Nnamdi Azikiwe University, Awka, Nigeria

2015 — 2019

Thesis: Geology and Environmental Geology Mapping of Ozara Ukwu, Ebonyi State

RESEARCH AND TECHNICAL EXPERIENCE

Co-Founder and Lead Developer | Verdant (EcoSim Platform)

May 2026 — Present

- Built a web-based ecological simulation platform (EcoSim) modelling degradation and restoration across 5 West African biomes using Three.js WebGL, Node.js, and custom GLSL shaders
- Implemented quantitative ecological health model governing real-time ecosystem state; paper submitted to Russia-Africa Youth Forum, RUDN University, 2026

Graduate Researcher | Institute of Environmental Engineering, RUDN University

2025 — 2026

- Mapped urban land cover change and flood vulnerability across all 20 LGAs of Lagos State using 6 satellite datasets (Landsat, Sentinel-2, GHSL, ESA WorldCover, SRTM, JRC) in Google Earth Engine
- Key findings: 41.1% urban expansion 2000-2020; 3.77M residents in high flood zones; drainage density 0.20 vs recommended 2.0 km/km²; Pearson r = -0.78 (imperviousness vs vegetation loss)

Full-Stack Software Engineer / DevOps | Mobr, Marseille, France (Remote)

Jul 2024 — Apr 2026

- Built RESTful APIs and admin dashboard in .NET Core and Angular; automated AWS CI/CD pipelines reducing deployment time by 35%

Environmental Research Consultant | Climate Technology Solutions

2023 — Present

- GIS analysis and community engagement for gully erosion remediation and sustainable land management across Anambra and Ebonyi States, Nigeria

Junior Geologist | Delta State Ministry of Water Resources, Asaba

Sep 2022 — Jul 2024

- Conducted hydrogeological field investigations across erosion-affected communities; produced technical reports informing water infrastructure investment decisions for rural populations

Backend Engineer | FlatPurse, Ontario, Canada (Remote)

Sep 2023 — May 2024

- Built real-time data streaming pipeline using Amazon Kinesis, Lambda, and QuickSight for a POS and e-commerce platform

Hydrogeology Trainee | Key Drilling and Geotechnical Nigeria

Apr 2020 — Oct 2021

- Borehole logging, aquifer testing, and geological mapping across erosion-prone communities in southeastern Nigeria

PUBLICATIONS AND PRESENTATIONS

Oriahki O., Kareem O., Ebite B., Adayi L.E., Okonkwo S.C. (2025). GIS-Based Waste Management Suitability Analysis. *Journal of Energy Technology and Environment*, 7(4), 44-52.

Ojefia F.E. et al., Okonkwo S. (2024). Geotechnical Investigation of Road Failure, Ogheghe, Nigeria. *Path of Science*, 10(9).

Oral presentations: XXXII Lomonosov International Conference, Moscow State University (Apr 2025, competitive selection); II International Youth Scientific Conference on AI Technologies in Science and Education (Dec 2024)

SKILLS AND CERTIFICATIONS

Geospatial: ArcGIS, QGIS, Google Earth Engine, Landsat/Sentinel-2, NDVI/NDWI, spatial statistics

Programming: Python (GeoPandas, Rasterio), C#/.NET Core, TypeScript, JavaScript, Node.js, SQL

Cloud/DevOps: AWS (Solutions Architect Associate, Data Engineer Associate), Docker, Kubernetes, Jenkins

Field Methods: Hydrogeological investigation, borehole logging, geological mapping, Environmental Impact Assessment